

73-103 · CMU-Q

Principles of Macroeconomics

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Measuring a Nation's Income

Wed 26 and Mon 31 Aug

Course Roadmap

	Topic	Description	Tools	Use
Measurement	GDP	What the economy produced	Circular flow, $Y \equiv C+I+G+NX$	Read a release; Qatar energy GDP
	Prices	Persistent rise in the price level	CPI / deflator, real vs nominal	Contract, wage, peg
	Labor	Employment, unemployment, participation	Labor-force identities; short AI/labor	GCC hiring / participation
Short-run demand	Goods / multiplier	Demand determines Y	ZZ, multiplier, Blanchard 3	Fiscal impulse
	Financial markets	How i is set	Money market, Blanchard 4	QCB / Fed; $i = \bar{i}$
	IS-LM	Goods + financial together	IS, LM, policy mix, dashboards	Shock → which curve, which way
Medium run	Phillips	Inflation and unemployment together	Phillips curve, Blanchard 7–8	Wage / inflation tradeoff
Long run	Growth	Why levels differ	Production function, growth accounting	Decomposing changes in productivity
Open	Open economy	Trade, rates, and the peg	NX, Blanchard 17–19	Shock → NX / peg (Qatar)

Two meetings on GDP

SESSION 1 · Wednesday 26 August

1

Wed · Discussion of readings

CORE 3.1–3.4. What stuck from the articles?

2

Wed · What macro does

Objectives, tools, and the cycle.

3

Wed · Measuring Y

Why we measure.
Circular flow. $C + I + G + NX$.

SESSION 2 · Monday 31 August · today

4

Mon · Nominal vs real

Qatar: the dollar total is not tonnes.

5

Mon · PPP

A haircut in Doha vs one in Manila.

6

Mon · Limits to GDP, Business cycle, quiz

What Y misses. Rule of 70. Quiz 1.

BLOCK 1

Discussion on readings

What did you take from the yen-carry article, Huang vs Amodei, and CORE 3.1–3.4?

After Monday

1 Yen-carry trade

What did the Foreign Policy article add to Monday's discussion on Yen carry trade?

2 Huang vs Amodei

After the video and the Axios piece, is your view of AI and jobs more nuanced, or more extreme?

3 CORE 3.1-3.4

What was the most interesting idea in the reading, and why?

BLOCK 2

Macroeconomics : Objectives and Scope

Objectives of Macroeconomic Policy and What does Macroeconomics Study

What are the objectives of macroeconomic policy?

Growth

Steady expansion without boom–bust cycles.

Full employment

Jobs for those who want to work, near the natural rate.

Redistribution

How the gains are shared.
Not the same object as GDP.

Price stability

Low, predictable inflation.
Preserves purchasing power.

Financial stability

A banking system that does not seize up.

Exchange-rate stability

Predictable relative prices for trade and capital.

Fiscal Policy

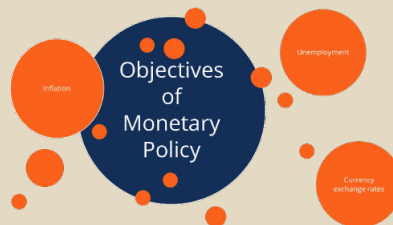
G and T. The government.

Monetary Policy

The policy rate, reserves, open-market operations.

Exchange rate Policy

Fix, float, or manage.



What does macroeconomics study?

1

Fluctuations

Expansions and contractions, the business cycle.

2

National output

The production of goods and services. GDP is the object.

3

Labor markets

Employment, unemployment, participation.

4

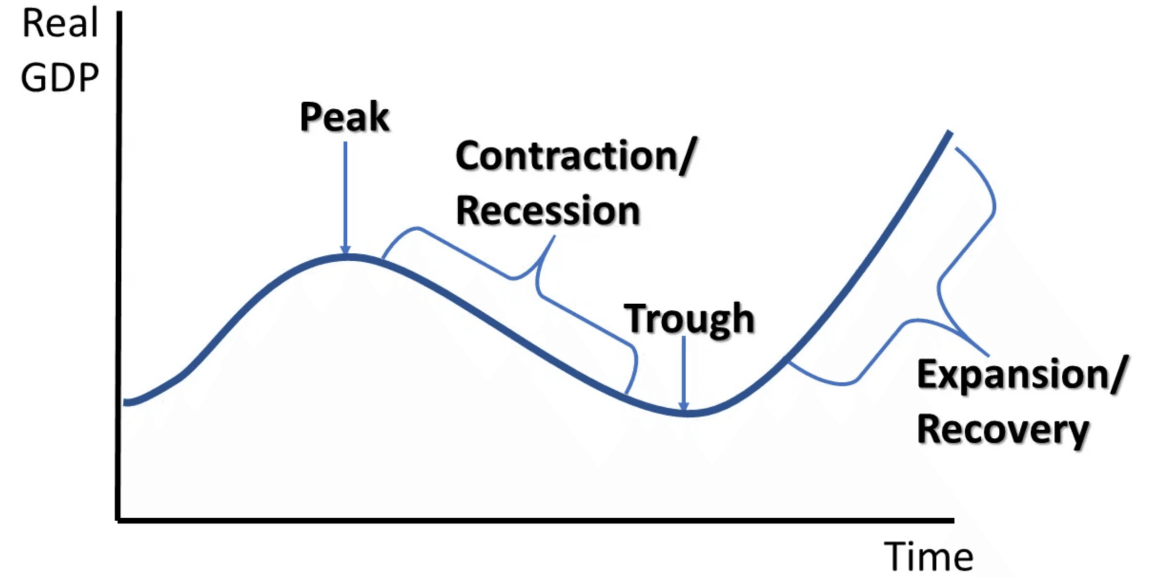
Price level

Inflation and deflation. Not one relative price.

5

Growth

The long-run path, distinct from the cycle.



The business cycle — peak, contraction, trough, expansion.

Where is the Qatari economy in this cycle today?

BLOCK 3

Why measuring economic conditions

Why does a firm, a household, and a finance ministry all need a measure of activity?

Why measuring activity matters

Prevailing conditions guide production, consumption, and public finance decision.

Firms

Produce

Production, inventories, hiring, and capex are set on a view of demand.

Households

Work · borrow

Whether to enter work, invest in schooling, or take on a mortgage.

Governments

Debt / GDP

taking more debt to expand social programs.

Reporting changes in activity is fundamental for decisions about production, consumption, and investment.

How changes in activity are reported

GDP

Production

The market value of output.
Today's object: the total
income of everyone in the
economy.

π

Inflation

The rate of change of the
price level.

u

Unemployment

The share of the labor force
without work.

NX

External balance

Net exports, and the
current account.

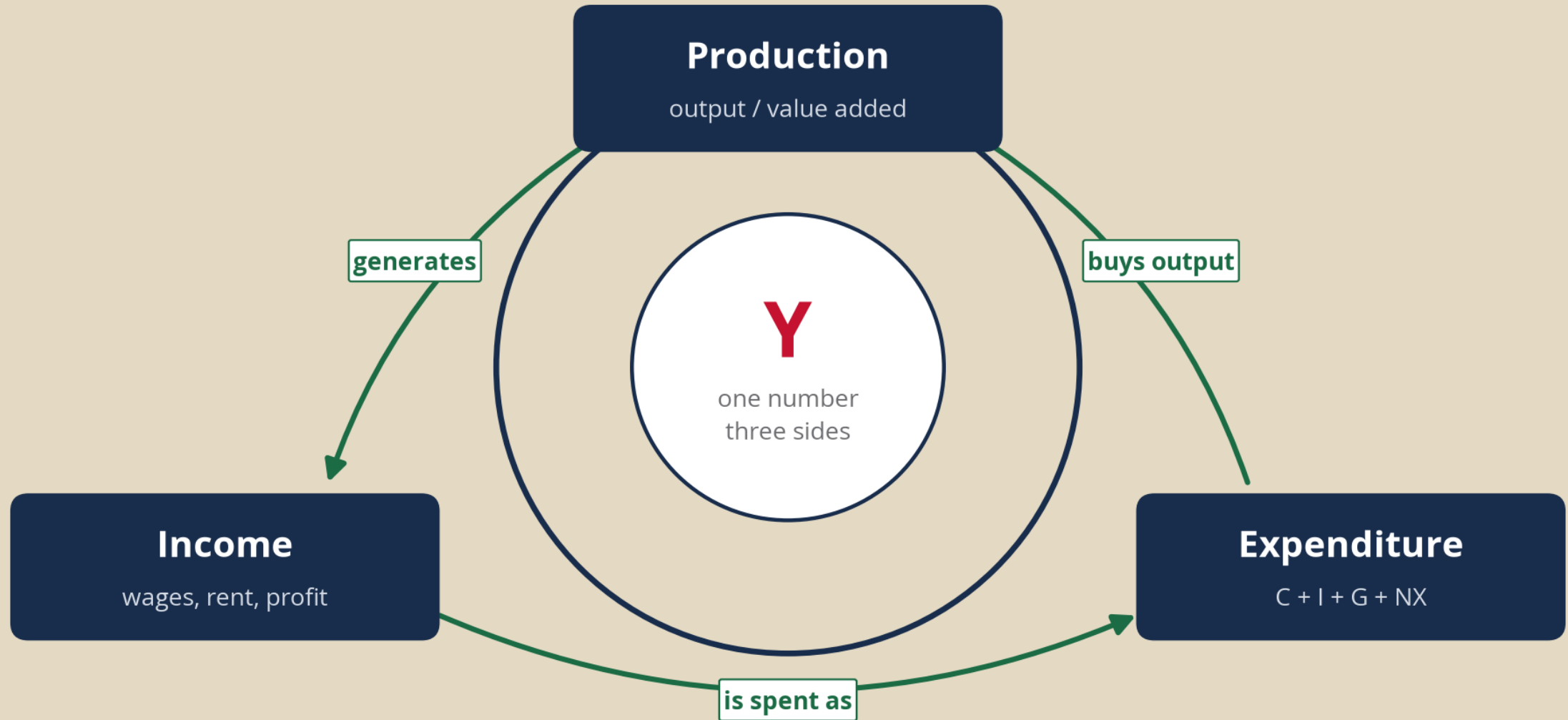
BLOCK 4

GDP Measurement and the Circular-Flow Diagram

Three (3) different ways to measure GDP

The circular flow model

The circular flow · production, income, and expenditure are the same Y



Three ways to measure GDP

If we counted everything, these are equal. Recessions: output falls, so expenditure and income fall.

Valued-added

$$Y = \sum VA_i$$

$$VA_i = P_i Q_i - \text{intermediates}_i$$

- Stops double-counting.
- The shirt, not cotton + cloth + shirt.

Income

$$Y = W + R + \text{Profits} + \text{others}$$

- Wages, rent, interest, and profit.
- What the owners of labor, land, and capital receive.

Expenditure

$$Y \equiv C + I + G + (X - M)$$

- True by construction.
- Inventories sit in I so output still equals spending when firms produce more than they sell.

Example: Producing a shirt, three industries

Value added at each stage = price of output – intermediate inputs



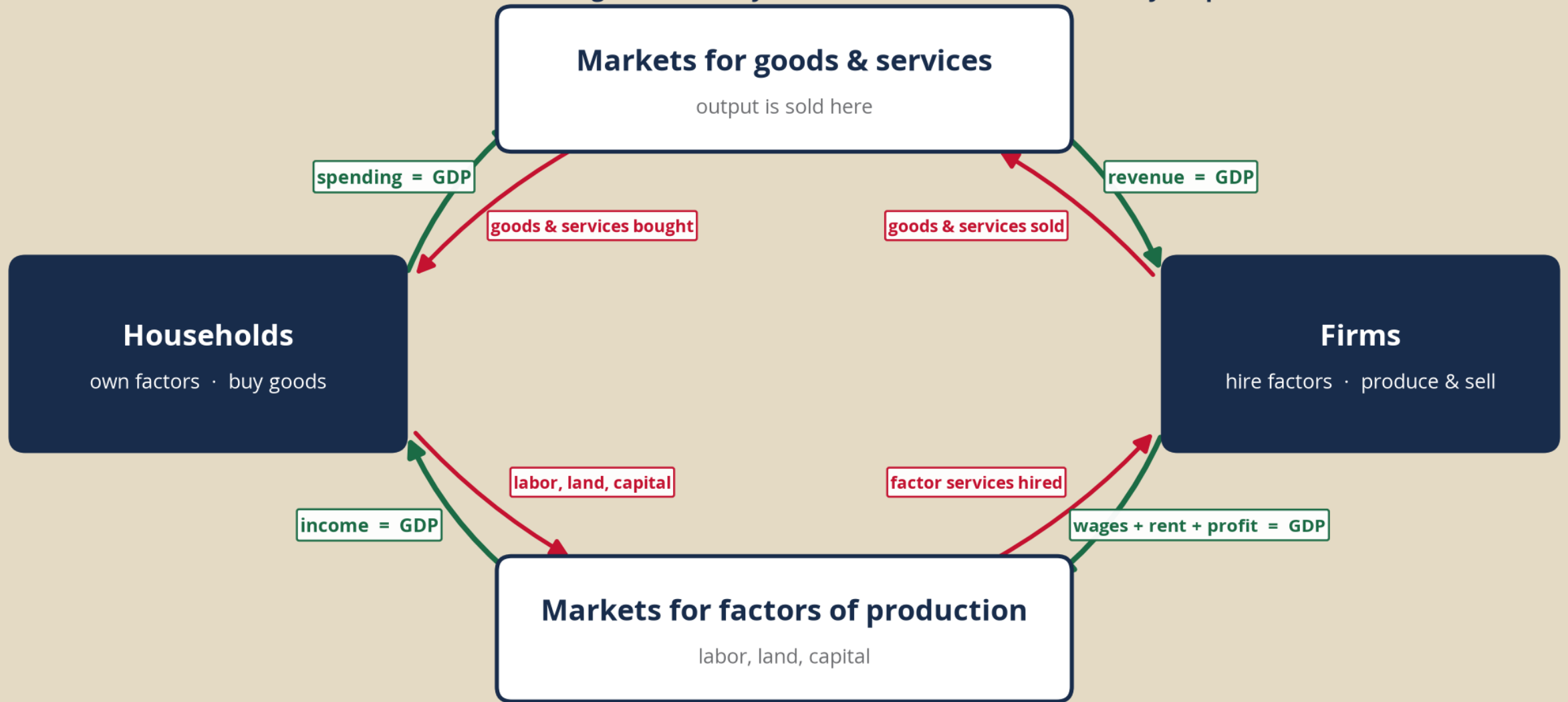
$$\text{GDP} = 50 + 30 + 20 = 100 = \text{final sale}$$

Wrong: $50 + 80 + 100 = 230$ (double-counting intermediates)

Same algebra for LNG: wellhead → liquefaction train → tanker. Do not add the three invoices.

Circular flow model: Two sectors

Two-sector circular flow • green = money • red = real • both money loops are GDP



What two sectors omit

Government

T

taxes

G

purchases

TR

transfers $\neq$
G

- Pensions and unemployment insurance are TR, not G.
- The retiree's spending of the check is already in C.

Financial system

S



I

- Saving finances investment.
- Buying a QNB share is a financial transaction, not I in the accounts.

Rest of the world

X

exports

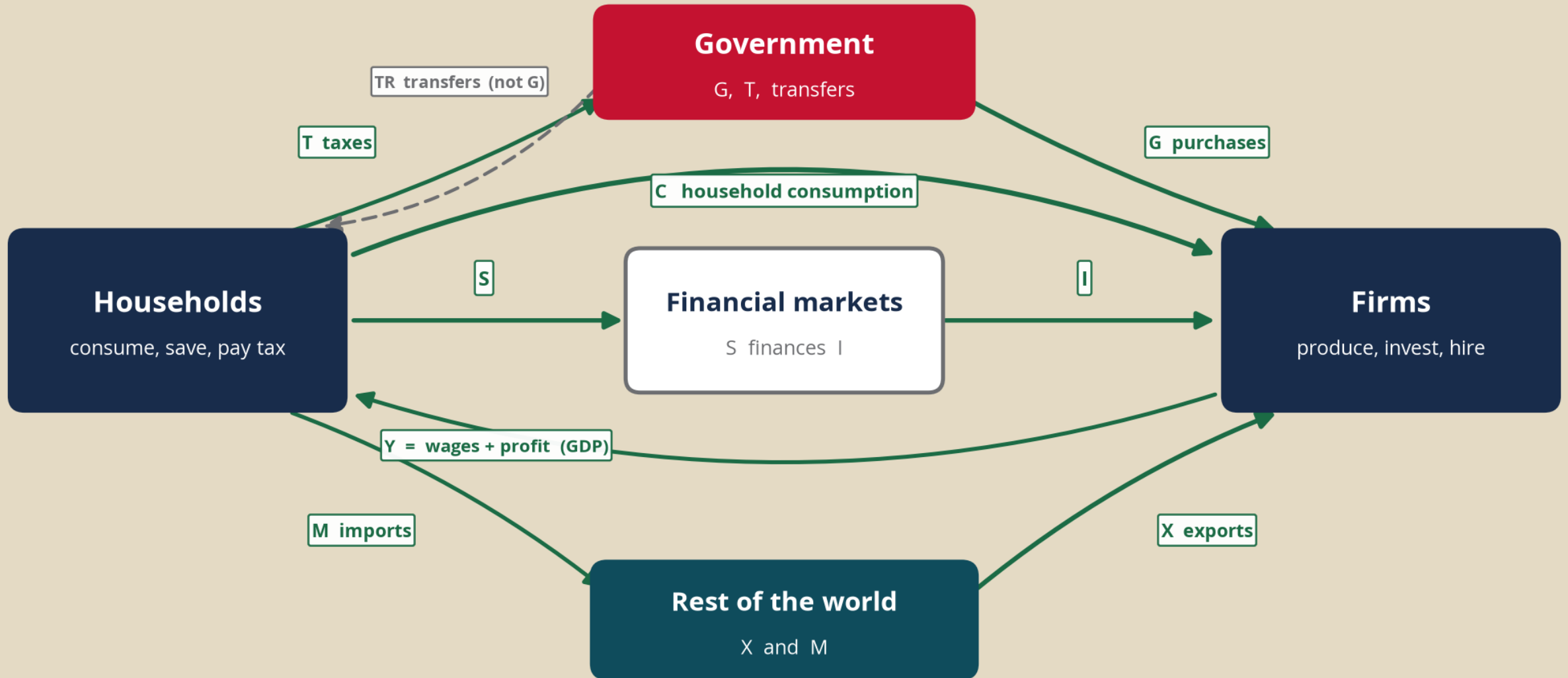
M

imports

- Net factor income is how GDP and GNI come apart. Trace X in, M out.

Four-sector circular flow

Four-sector circular flow — money. Real goods and factors run the opposite way. Dashed = transfers (not G).



BLOCK 5

The components of GDP

GDP: Components of the expenditure approach

The Components of GDP

C

Consumption

Spending by households on goods & services

I

Investment

- Purchase of capital goods, structures by household and firms, and
- inventories
- Also called gross fixed capital formation

G

Government

Government purchases of goods & services

NX

Net Exports

Exports minus Imports (X – M)

$$Y = C + I + G + NX$$

Consumption (C)

Household spending on goods and services. New housing is I, not C.

Durables

Cars, appliances. Last more than a year.

Nondurables

Food, fuel, clothing.

Services

Haircuts, health, education.

Housing

Renter

Rent paid is C.

Owner

Imputed rent is C. Not the purchase price, not the mortgage.

New house

The structure is I, not C.

Investment (I)

Spending on goods that will be used to produce more goods. Also called gross fixed capital formation.

Business capital

Structures, equipment, intellectual property.

Residential

A landlord's building; a homeowner's residence.

Inventories

Goods produced but not yet sold.

Not I

Buying QNB shares or a Treasury bill is a financial transaction. It is not investment in the national accounts.

Government purchases (G)

G = purchases

Goods and services

Teachers, hospitals, the metro, a new school.

Federal, state, and local.

Government consumption expenditure and gross investment.

Transfers $\neq$ G

Not a purchase

Social Security, unemployment insurance.

Double-count if you add them

The retiree's spending of the check is already in C.

G is taken as given this period.

Net exports (NX)

X

Foreign spending on our goods and services.

—

M

The parts of C, I, and G spent on foreign output.

=

NX

X - M

Adding the components

$$Y = C + I + G + NX$$

Imports already sit inside C, I, and G. Subtracting M stops us counting foreign output as ours.

Two Very Different Economies

United States

Consumption (C)	68%
Investment (I)	18%
Government (G)	17%
Net Exports (NX)	-3%

A consumption-driven economy with a trade deficit.

Qatar

Consumption (C)	22%
Investment (I)	38%
Government (G)	12%
Net Exports (NX)	+28%

An export-driven economy (LNG/Oil) with a large trade surplus.

BLOCK 6

Active Learning

QatarEnergy import modules for North Field train

QatarEnergy: modules built in Korea QR 800; installation in Qatar QR 50. What happens to I, NX, and Y?

ΔC	ΔI	ΔG	ΔNX	ΔY
0	+ QR 850	0	- QR 800	+ QR 50

$$0 + 850 + 0 - 800 = 50 \quad (\text{QR})$$

$$I = 800 + 50$$

Fixed capital on the territory: imported modules plus local installation.

$$NX = -800$$

The modules are M. Korea's factory is Korea's GDP.

$$Y = 50$$

Only installation VA is produced here. Ownership of QatarEnergy does not matter.

A Doha household buys an imported iPhone

Retail price QR 4,000. Import cost QR 3,500. Local retailer's margin QR 500. What happens to C, NX, and Y?

ΔC	ΔI	ΔG	ΔNX	ΔY
+ QR 4,000	0	0	- QR 3,500	+ QR 500

$$4,000 + 0 + 0 - 3,500 = 500 \quad (\text{QR})$$

$$C = 4,000$$

The household's purchase is consumption at the retail price.

$$M = 3,500$$

The foreign factory's value leaves through imports. NX falls by 3,500.

$$Y = 500$$

GDP is the retailer's margin, domestic value added. A tourist buying the same phone would be X of the retail service, not C.

Active Learning 1: Is it in this year GDP?

A Lusail stadium, built in Qatar in 2019 by a local contractor.

B You resell a 2022 World Cup ticket to a visitor from London.

C QatarEnergy pays for a new LNG train on the North Field. Modules are built in Korea.

D A 2019 Land Cruiser, listed on Qatar Living.

Is it in this year GDP? — mechanism

A • no

New structure on the territory.

Investment, and GDP, in 2019.

Not this year's GDP.

B • no

A resale is not current production. The original ticket, if sold by FIFA/Qatar, was already counted.

C • I and -M

Fixed capital on the territory is I. Imported modules are M. GDP rises only by value added in Qatar (installation).

D • no

Used good. Production was in an earlier year. A dealer's margin, if any, would be current value added.

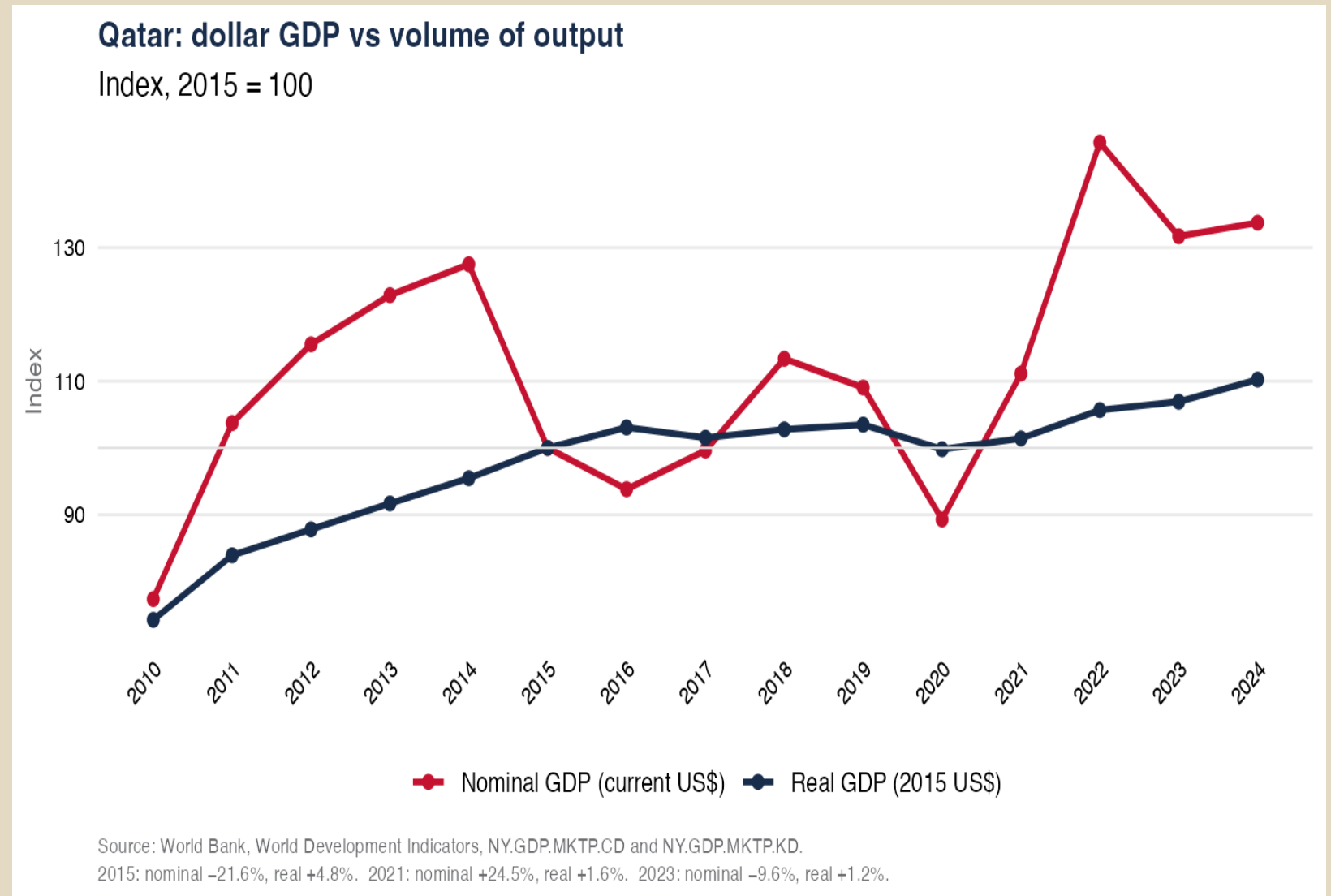
SESSION 2

Nominal, real, and PPP- adjusted GDP

How we can compare incomes at different times, and across different countries.

Motivation: Evolution of Qatar GDP

- What do you see in this graph?
- Which line tell a more compelling story about the production of the Qatar's economy?



NOMINAL VS REAL

Nominal vs Real GDP

If total spending rises, is it more stuff, higher prices, or both?

If nominal GDP rose, which (P, Q) case?

Nominal GDP is $P \times Q$. Five ways that product can rise.

1 Q ↑ , P still	2 P ↑ , Q still	3 Both ↑	4 P ↑ faster than Q ↓	5 Q ↑ faster than P ↓
More stuff at the same prices.	Same stuff, higher prices.	Volume and prices both rose.	Inflation masks a quantity fall.	Volume up even if some prices fell.

Why split the two effects?

Real GDP values output at constant (base-year) prices so we can compare quantities.

Nominal GDP

The value of production at current prices.

Formula

$$\text{Nominal GDP}_t = \sum (P_t \times Q_t)$$

- Current price $\times$ current quantity.
- For the base year, nominal = real.

Use

- The dollar value of transactions this year.
- Debt/GDP,
- tax take, financial flows.

The limitation

10% more stuff?

Growth in Q .

10% higher prices?

Inflation in P .

A mix?

You cannot tell from the nominal value alone.

Real GDP

The value of production at constant (base-year) prices. Isolates Q.

Formula

$$RealGDP_t = \Sigma(P_{base} \times Q_t)$$

- Hold prices still; let quantities move.
- For the base year, real = nominal.

Use

- Growth and the size of the economy over time.
- Qatar 2022-23: real still rose when the nominal GDP fell.

Real versus Nominal GDP

Nominal

- **Nominal GDP = $\Sigma (P_t \times Q_t)$**
- Current prices.
- Not corrected for inflation.
- The print for debt, taxes, and cash flows.

Real

- **real GDP = $\Sigma (P_{\text{base}} \times Q_t)$**
- Base-year prices.
- Corrected for inflation.
- The print for growth and volume.
- Base year: nominal = real.

PPP GDP

Comparing across countries

A haircut in Doha is not a haircut in Manila.

When real GDP is not enough

What can you actually buy with the same \$100?

New York

\$100

2 movie tickets + popcorn



Manila

\$100

20 movie tickets + 10 meals



Doha

\$100

1 fancy dinner, no appetizers



Challenge: how do we compare output when prices vary this much across countries?

A haircut in Doha and Manila

Doha

\$30



VS

Manila

\$3



- If we just convert currencies, the Doha economy looks 10x larger for the same output. Is this comparison ok?
- We need a measure that accounts for these **Cost of Living** differences.

PPP-adjusted GDP

Why not just use the market exchange rate?

Volatile

The FX rate moves daily. A living-standard comparison should not.

Price levels

A haircut costs less in Manila than in Doha. FX ignores that.

Solution

PPP GDP = Nominal GDP (local currency) / PPP conversion factor

Example: Classroom chicken index

- 1 kg chicken:
 - 100 INR in India,
 - \$5 in the US.
 - PPP rate = $100 / 5 = 20$ INR per international \$.
 - The market rate might be ~83 INR/USD.

How PPPs are calculated it ?

1. A common basket.
2. Local-currency prices.
3. A weighted ratio: the conversion factor.

Case study: Qatar vs Philippines

Measure	Philippines	Qatar
Total nominal GDP	\$404 bn	\$237 bn
GDP per capita (USD)	\$3,500	\$82,000
GDP per capita (PPP)	\$10,000	\$115,000

Size vs wealth

- The Philippines is larger in total because 115 million people vs about 3 million.
- Per-person, Qatar is far richer.

The PPP effect

- Philippine income roughly triples (\$3.5k → \$10k) at PPP.
- Manila's price level is lower than global prices.

What each measure tells us

Measure	Best used for...
Nominal GDP Current prices	Debt ratios, tax take, international cash flows.
Real GDP Constant prices	Growth over time. Removes inflation effect.
PPP-adjusted GDP Common basket	Living standards across countries. Removes price-level differences between countries.

Rule of thumb: compare time with real. Compare places with PPP

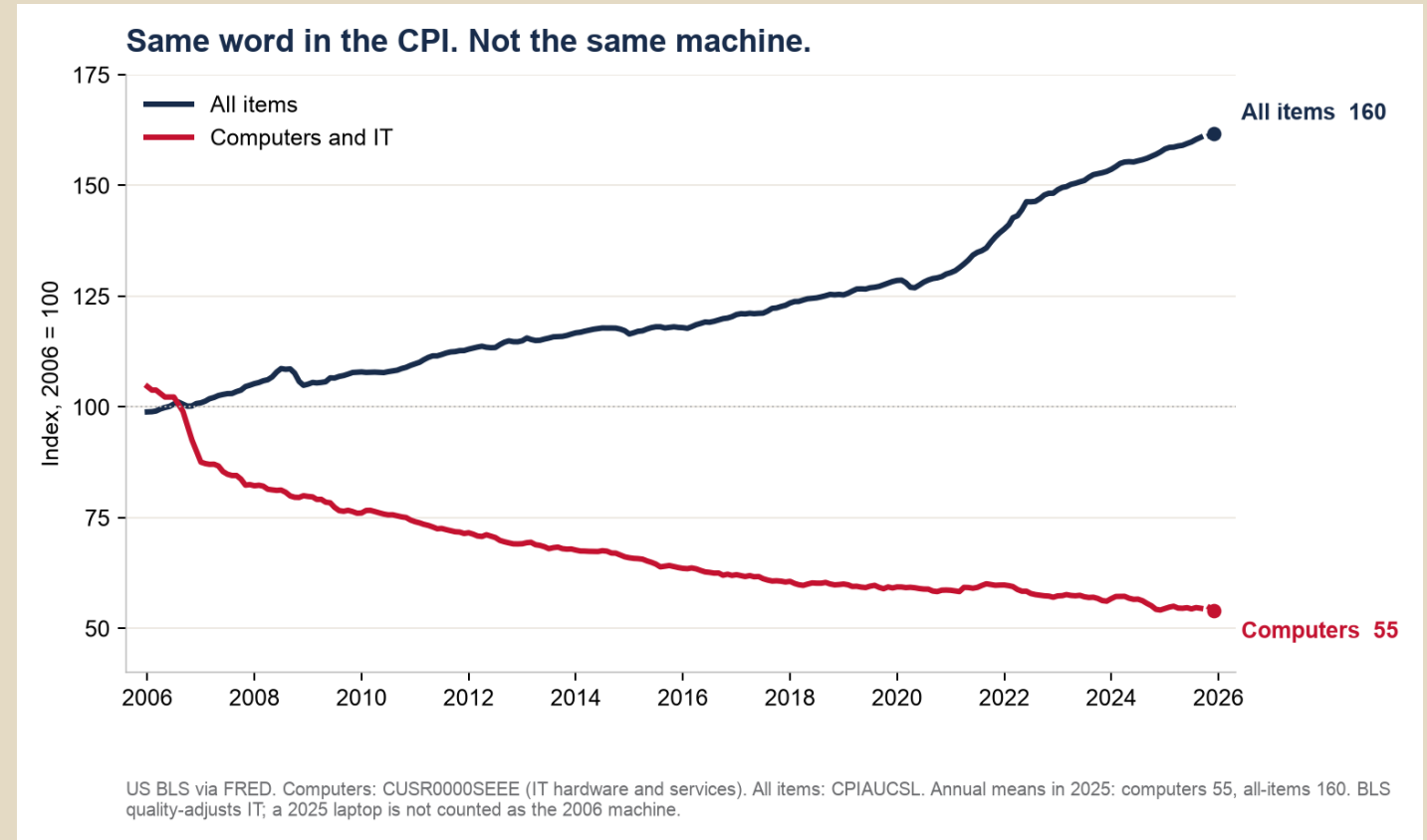
WHAT Y MISSES

Three holes in the print

Limitations of the GDP

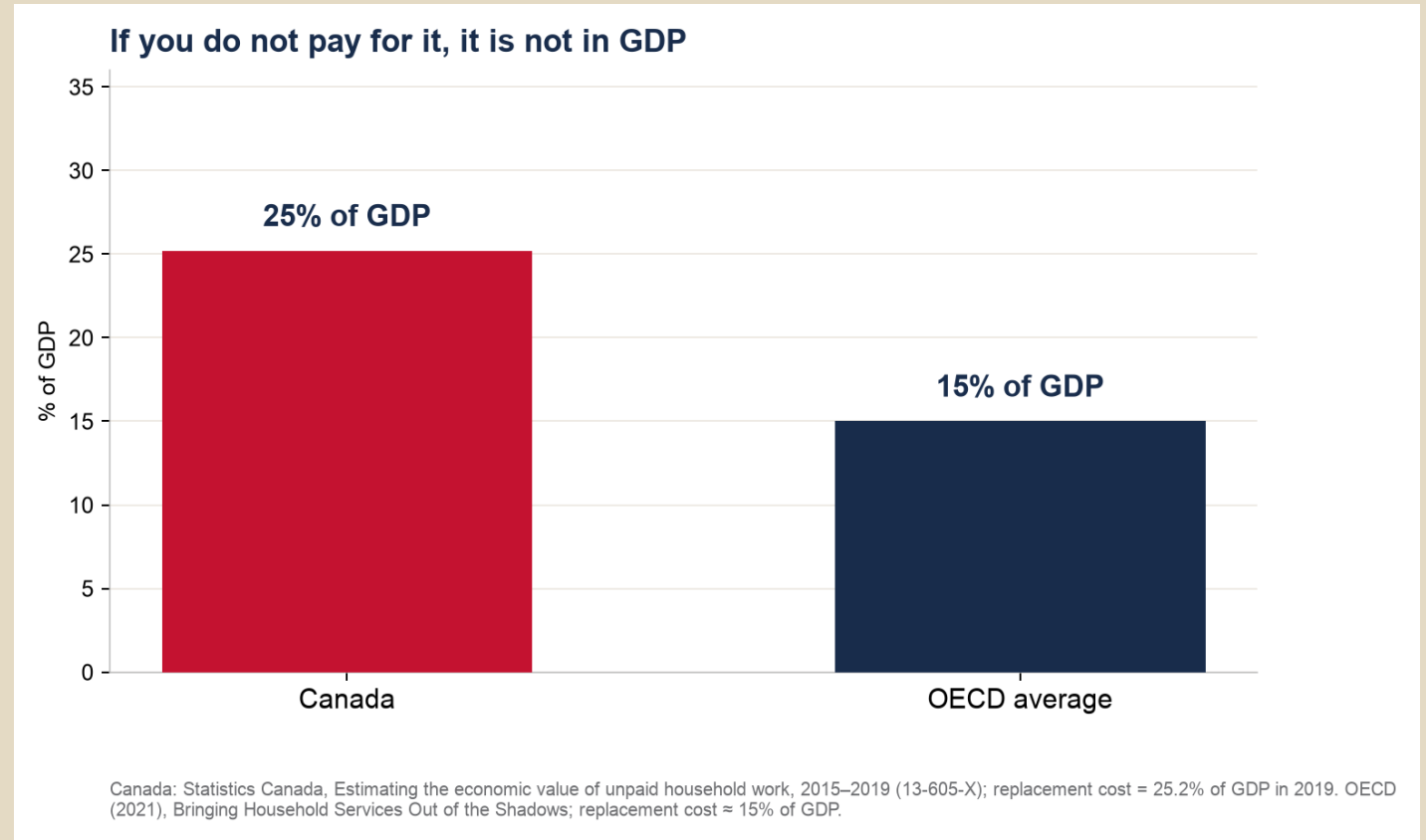
Limitation 1 — Quality change

- A 2024 laptop is not a 2006 PC with a new coat of paint. You bought more computing.
- If the accounts treat them as the same “computer”, real IT output looks too low. BLS quality-adjusts. That is why the computer line falls while the rest of the CPI rises.
- By 2025 the computer CPI is about 45% below its 2006 level. All-items is about 60% above.



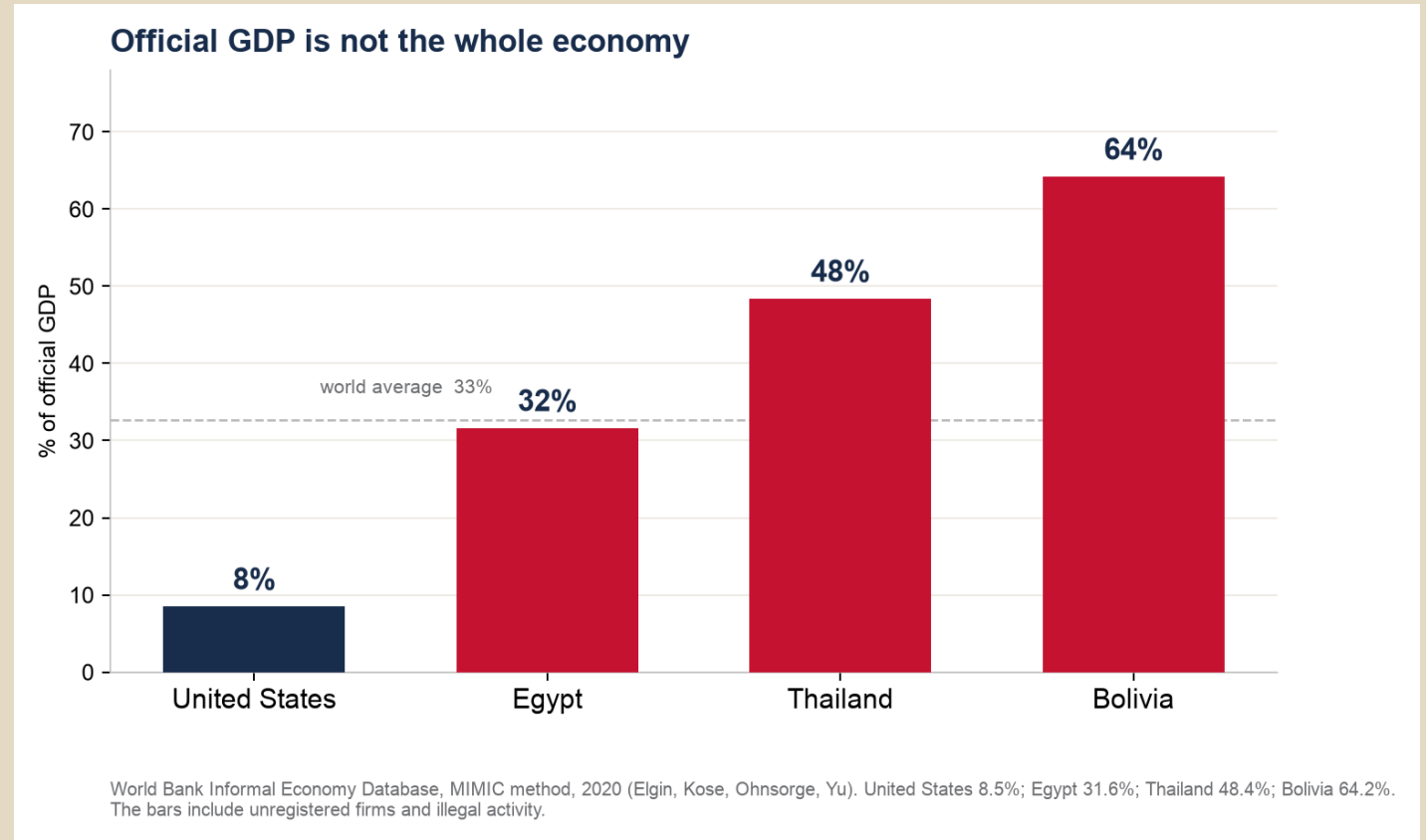
Limitation 2 — unpaid household work

- Cook dinner at home: GDP does not move. Pay a restaurant for the same meal: GDP rises.
- Unpaid cooking, cleaning, and care are production. They are not sold, so they never enter Y.
- Statistics Canada puts that work at about a quarter of GDP if you value it at what it would cost to hire.
 - The OECD average, same method, is about 15%.



Limitation 3 — unregistered output

- A street stall and an unregistered workshop are production. They often never make the books.
- In the United States that gap is about 8% of official GDP. In Egypt, about one-third. In Thailand, about half. In Bolivia, closer to two-thirds.
- Illegal activity is inside these bars. Most of the gap is ordinary unregistered work.



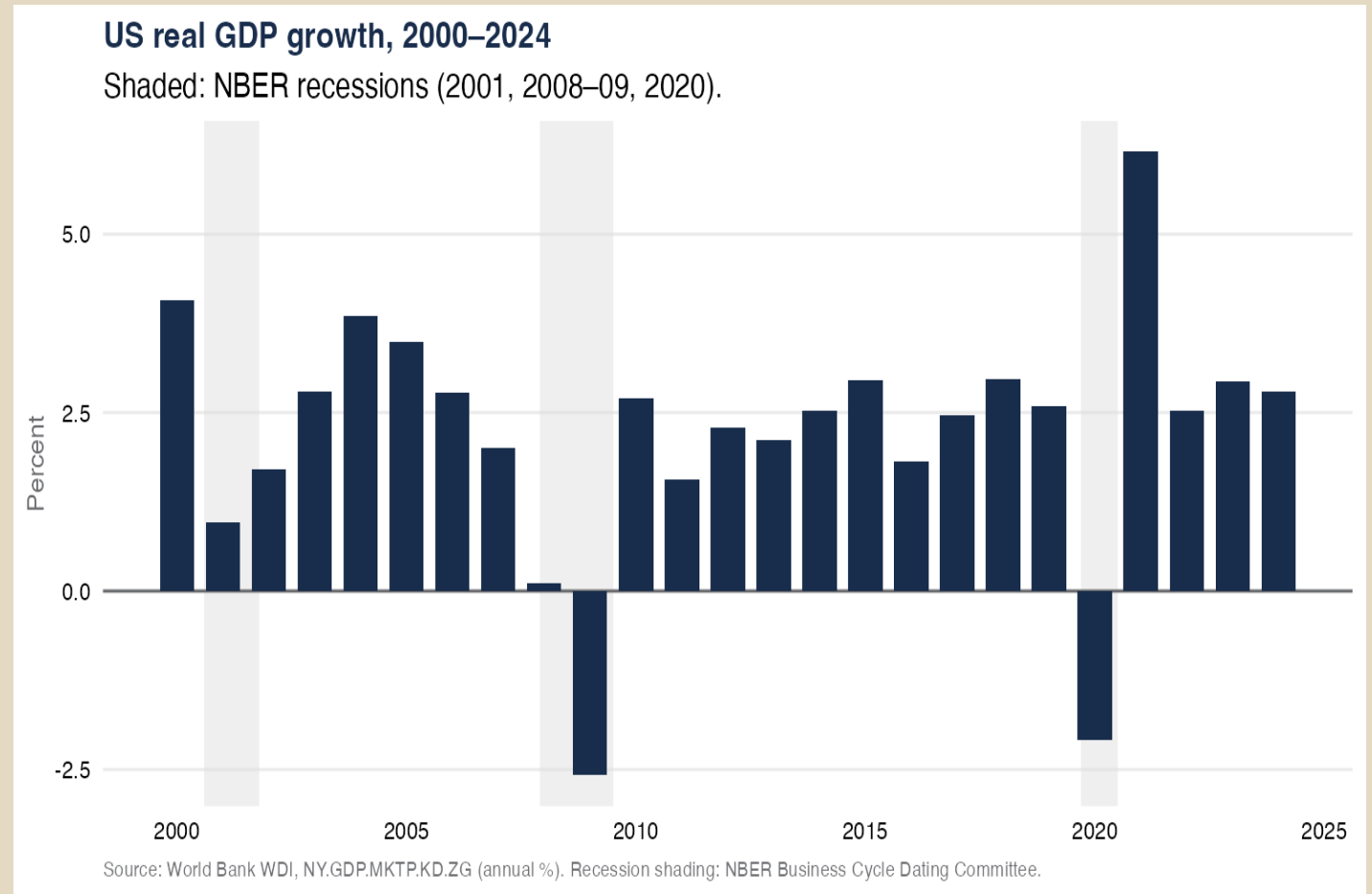
GROWTH AND THE CYCLE

**Growth is not a
smooth path**

**Business Cycle and the rule of 70 for GDP
growth**

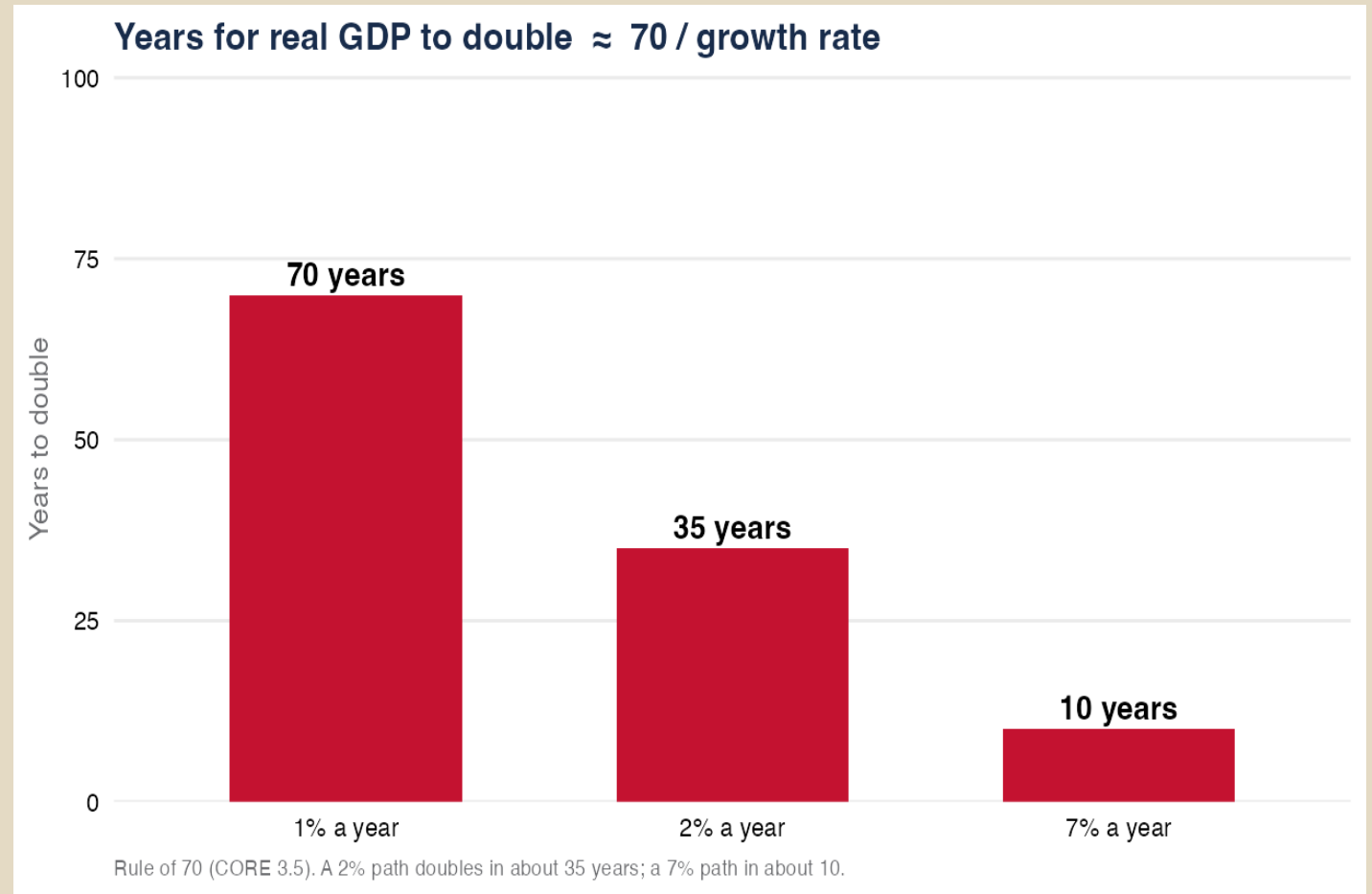
The business cycle

- Growth is not a smooth path.
- Recession definitions:
 - significant decline in activity, spread across the economy → two consecutive quarters
 - output below its normal level, even if it is growing again.
- Examples:
 - US: 2001, 2008–09, 2020.



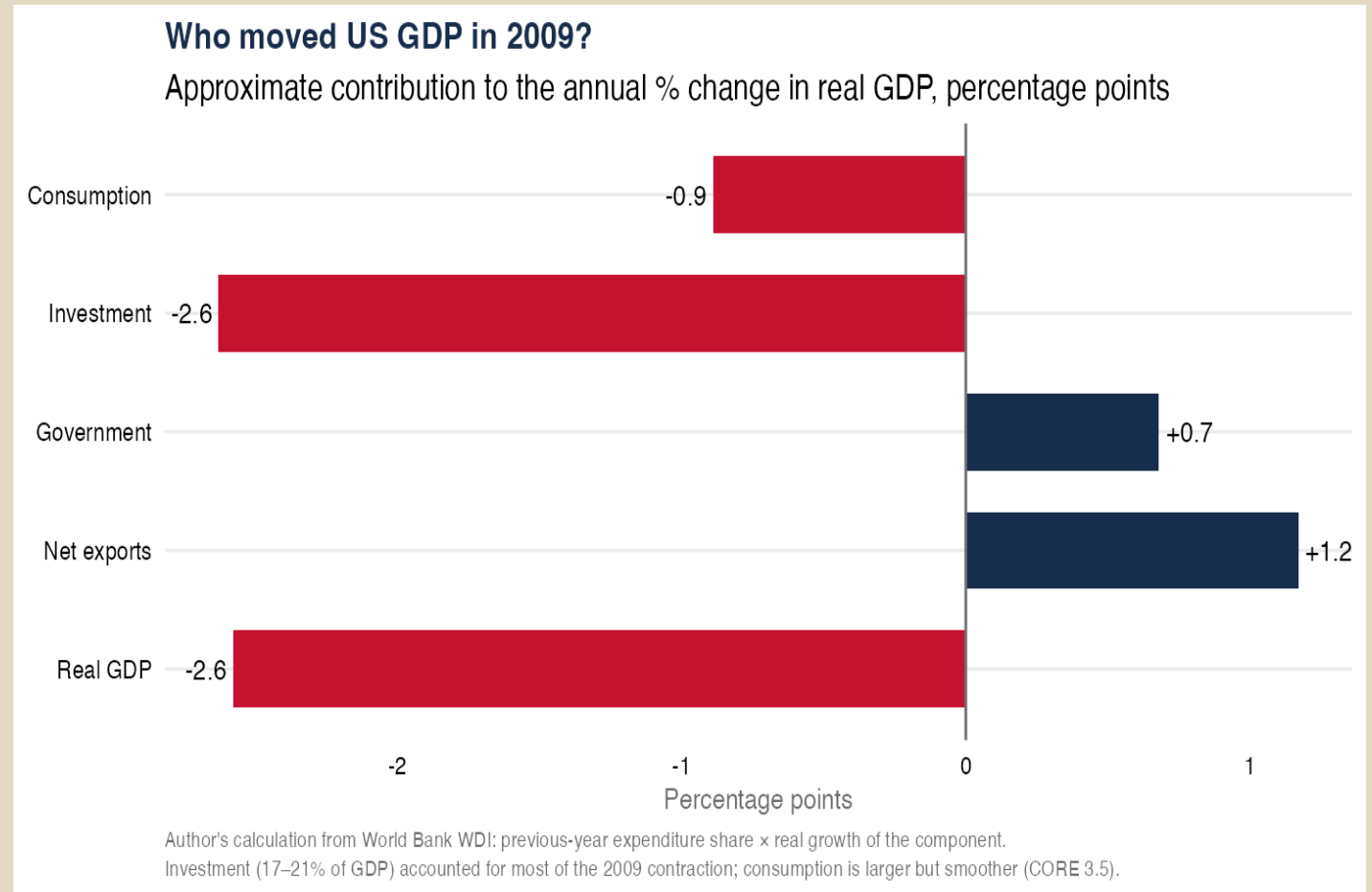
The rule of 70 for GDP growth

- *Years to double $\approx 70 /$ annual growth rate.*
- 2% a year $\rightarrow$ about 35 years.
- 7% a year $\rightarrow$ about 10.
- Think in terms of investing your own money in an asset that gives you an inflation adjusted return between 8-10% a year
 - What's that asset?



Who moved GDP?

- $\Delta Y \approx s_C \Delta C + s_I \Delta I + s_G \Delta G + s_{NX} \Delta NX$.
- A small share can still dominate the change if it is volatile.
- 2009: investment is ~20% of US GDP and accounted for most of the fall.
- Consumption is larger and smoother. That is why recessions hit I first.



CLOSE

Final takeaways about GDP

Is GDP a good measure of economic well-being?

Is GDP a good measure of well-being?

Robert Kennedy, University of Kansas, 18 March 1968:

“GDP does not allow for the health of our children, the quality of their education, or the joy of their play. It does not include the beauty of our poetry or the strength of our marriages, the intelligence of our public debate or the integrity of our officials. It measures neither our courage, nor our wisdom, nor our devotion to our country. It measures everything, in short, except that which makes life worthwhile, and it can tell us everything about America except why we are proud that we are Americans.”

However.....

APPENDIX

CALCULATIONS

Calculating GDP Indicators

EXAMPLE 1A: Calculating nominal GDP

Compute total spending (nominal GDP) in each year

Year	Baklavas P	Baklavas Q	Umm Ali P	Umm Ali Q
2019	\$15	300	\$2.50	1,000
2020	\$16	400	\$3.00	1,200
2021	\$17	500	\$3.50	1,300

2019: $\$15 \times 300 + \$2.50 \times 1,000 = \$7,000$

2020: $\$16 \times 400 + \$3.00 \times 1,200 = \$10,000$

2021: $\$17 \times 500 + \$3.50 \times 1,300 = \$13,050$

Increase

42.9%

30.5%

EXAMPLE 1B: Calculating real GDP

Compute real GDP in each year (base year 2019)

Year	Baklavas P	Baklavas Q	Umm Ali P	Umm Ali Q
2019	\$15	300	\$2.50	1,000
2020	\$16	400	\$3.00	1,200
2021	\$17	500	\$3.50	1,300

Use 2019 prices: \$15 and \$2.50

2019: $\$15 \times 300 + \$2.50 \times 1,000 = \$7,000$

2020: $\$15 \times 400 + \$2.50 \times 1,200 = \$9,000$

2021: $\$15 \times 500 + \$2.50 \times 1,300 = \$10,750$

Increase

28.6%

19.4%

EXAMPLE 1C: Correcting for inflation

Year	Nominal GDP	Real GDP
2019	\$7,000	\$7,000
2020	\$10,000	\$9,000
2021	\$13,050	\$10,750

Nominal increase

42.9%

30.5%

Real increase

28.6%

19.4%

The change in nominal GDP reflects both prices and quantities. The change in real GDP is the amount that GDP would change if prices were constant (i.e., if zero inflation). Hence, real GDP is corrected for inflation.

THE DEFLATOR

The GDP Deflator

How do we take the inflation out of nominal GDP?

The GDP Deflator

- The GDP Deflator gets its name because it is used to “deflate” (take the inflation out of) nominal GDP to get real GDP.

$$\text{GDP Deflator} = (\text{Nominal GDP} / \text{Real GDP}) \times 100$$

- For the base year, the GDP deflator is 100.
- The inflation rate is a percentage change in the price level: $100 \times (\text{new} - \text{old}) / \text{old}$.
- “New” is the new price level; “old” is the previous price level.

EXAMPLE 1: Calculate the GDP deflator

Compute the GDP deflator in each year

Year	Nominal GDP	Real GDP	GDP deflator	Inflation rate
2019	\$7,000	\$7,000	100.0	—
2020	\$10,000	\$9,000	111.1	11.1%
2021	\$13,050	\$10,750	121.4	9.3%

2019: $100 \times (7,000 / 7,000) = 100$

2020: $100 \times (10,000 / 9,000) = 111.1$

2021: $100 \times (13,050 / 10,750) = 121.4$

ACTIVE LEARNING 4

Calculating GDP Indicators

Computing GDP

	2019 P	2019 Q	2020 P	2020 Q	2021 P	2021 Q
Cookies	\$1	900	\$2	1,000	\$3	1,250
Smartphones	\$900	185	\$1,000	200	\$1,200	210

2019 is the base year. Use the data to:

Compute nominal GDP in 2019.

Compute real GDP in 2020.

Compute the GDP deflator in 2021.

Active Learning: Answers A, B

	2019 P	2019 Q	2020 P	2020 Q	2021 P	2021 Q
Cookies	\$1	900	\$2	1,000	\$3	1,250
Smartphones	\$900	185	\$1,000	200	\$1,200	210

A · Nominal GDP in 2019

$$\$1 \times 900 + \$900 \times 185 = \$167,400$$

B · Real GDP in 2020

$$\$1 \times 1,000 + \$900 \times 200 = \$181,000$$

Nominal: current quantities and current prices. Real: current quantities; base-year prices.

Active Learning: Answers, C

	2019 P	2019 Q	2020 P	2020 Q	2021 P	2021 Q
Cookies	\$1	900	\$2	1,000	\$3	1,250
Smartphones	\$900	185	\$1,000	200	\$1,200	210

C · GDP deflator in 2021

$$\text{Nom GDP} = \$3 \times 1,250 + \$1,200 \times 210 = \$255,750$$

$$\text{Real GDP} = \$1 \times 1,250 + \$900 \times 210 = \$190,250$$

$$\text{GDP deflator} = 100 \times (\text{Nom GDP}) / (\text{Real GDP}) = 100 \times 255,750 / 190,250 = 134.4$$

APPENDIX

The GDP WHOLE definition

What is GDP? (Part 1: Value & Scope)

GDP is the market value of all final goods & services produced within a country in a given period of time.

Market value

Goods are valued at market prices, so apples and cars can be added. Non-market activity (housework, most home production) is out.

Of all

Legal market production. Out: illicit transactions, and items produced and consumed at home.

Final goods

Intended for the end user. Intermediate goods are out — their value is already in the final good. That is the value-added lesson.

What is GDP? (Part 2: Nature, Location & Time)

GDP is the market value of all final goods & services produced within a country in a given period of time.

Goods & services

Tangible goods: food, clothing, cars. Intangible services: haircuts, concerts, doctor visits.

Produced

Only currently produced goods. A used car or an old house is not this period's GDP. A dealer's margin is.

Within a country

Production inside the border, regardless of the producer's nationality. A foreign factory in Qatar is Qatar's GDP.

Given period

A flow: usually a year or a quarter. Not a stock of wealth.

APPENDIX

How to model each component of GDP?

Consumption

$$C = C_0 + c_1 (Y - T)$$

$Y - T$

Disposable income.

$$C_0 \geq 0$$

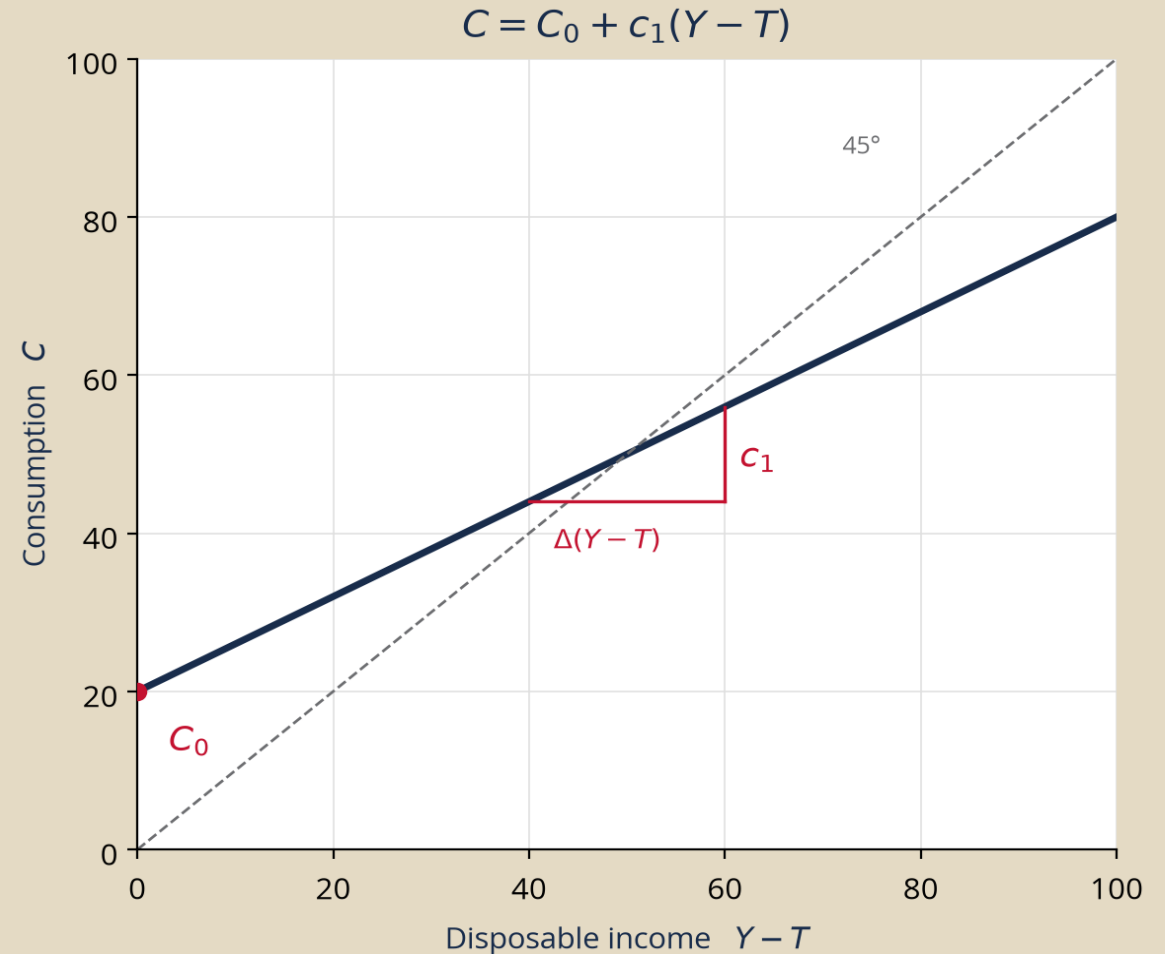
Autonomous consumption: what households spend even at zero disposable income (borrowing, drawing down assets).

$$0 < c_1 < 1$$

Marginal propensity to consume. When income rises, households spend part and save part. That split is goods-market week.

Not C

New housing is I, not C. Imputed rent on an owner-occupied house is C. The purchase price is not.



Investment

$$I = I_0 - b i$$

I_0

Autonomous investment: animal spirits, technology, North Field capex.

i interest rate; $b > 0$

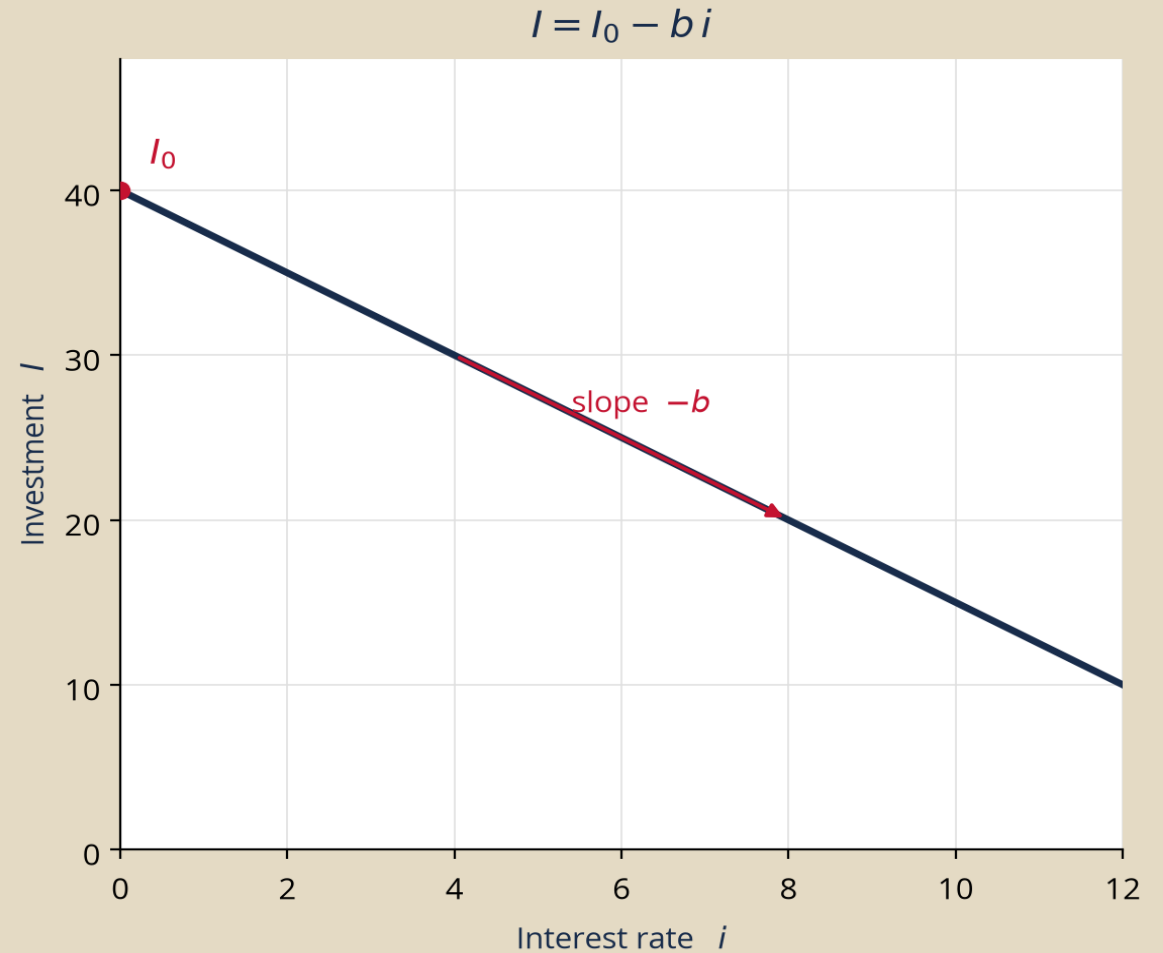
A higher i makes it more expensive to finance machines, structures, and housing. I falls.

In the accounts

Business capital (structures, equipment, IP). Residential structures. Inventory accumulation.

Not I

Buying QNB shares or a Treasury bill is a financial transaction. It is not I .



Government purchases

$$G = \bar{G}$$

Taken as given this period

Chosen by the fiscal authority, not by a household first-order condition. G is purchases of goods and services: teachers, hospitals, the metro, a new school.

The longer NIPA label is “government consumption expenditure and gross investment.”

Transfers are not G

Pensions, unemployment insurance, and social assistance are TR, not G .

The retiree’s spending of the check is already in C . Counting the check as G would double-count.

$\bar{G}$ can change next period. Today we do not endogenize it.

Net exports

$$NX = X(Y^*, \varepsilon) - IM(Y, \varepsilon)$$

$X(Y^*, \varepsilon)$

Exports rise with foreign income Y^* . They also depend on the real exchange rate ε .

$IM(Y, \varepsilon)$

Imports rise with domestic income Y : richer households and firms buy more foreign goods.

ε , Blanchard's convention

$\varepsilon = EP/P^*$ — price of domestic goods in terms of foreign goods. A real appreciation ($\varepsilon \uparrow$) reduces NX .

Compact form, later

$NX = NX_0 - nY$, given Y^* and ε . $n > 0$.

In the accounts

A Doha hotel night sold to a visitor is X . An iPhone assembled abroad sits in C and in IM .

